



Predicting H1N1 Vaccination Uptake

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Overview of The Task

We want to understand **who gets vaccinated and why**

Why it matters:

- ▶ To improve public health campaigns
- ▶ To prepare for future pandemics
- ▶ To target resources effectively

The Data

Source: 26,000+ respondents from the 2009 National H1N1 Flu Survey

What was looked at:

- ▶ **Opinions:** Is the vaccine safe? Am I at risk?
- ▶ **Behaviours:** Do they wear masks? Wash hands?
- ▶ **Demographics:** Age, income, race, and employment.

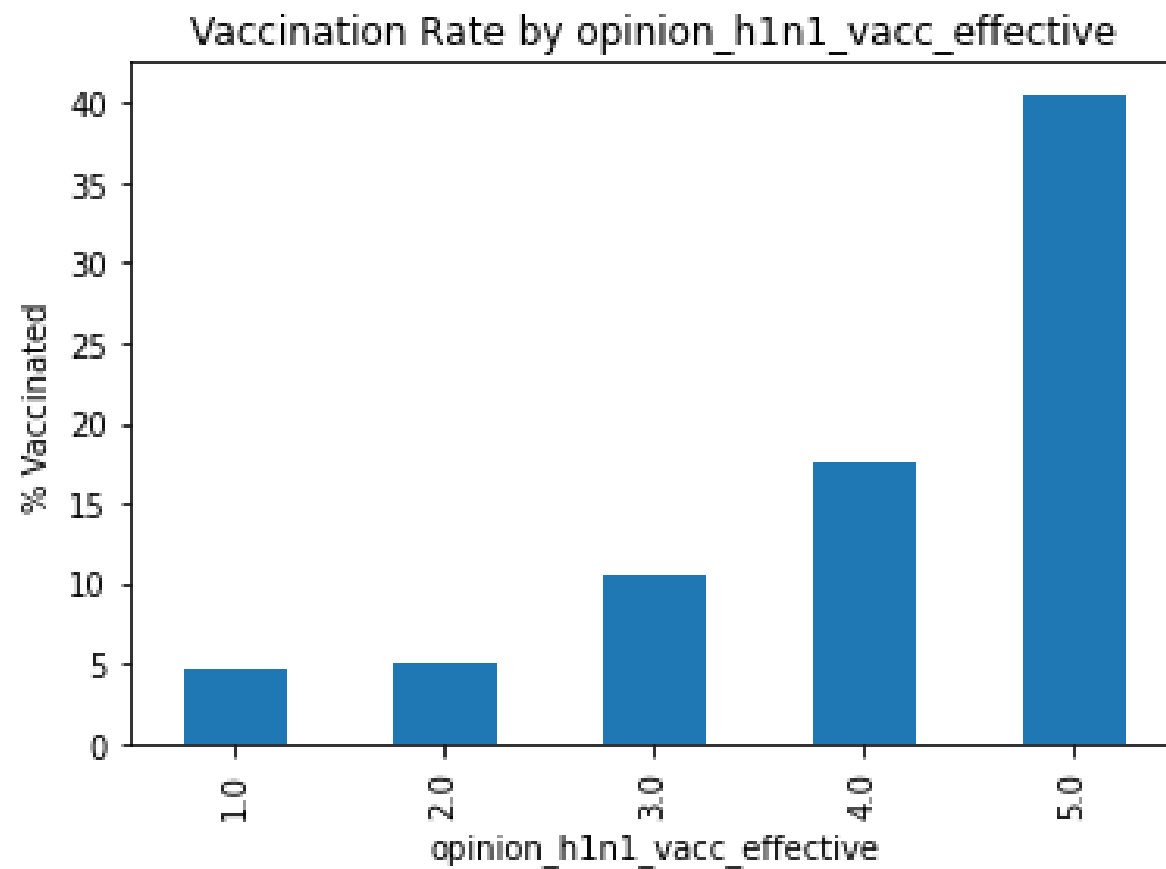
Data Handling

- ▶ EDA
- ▶ Visualization
- ▶ Cleaning
- ▶ Imbalance fixing
- ▶ Modeling
- ▶ Model evaluation

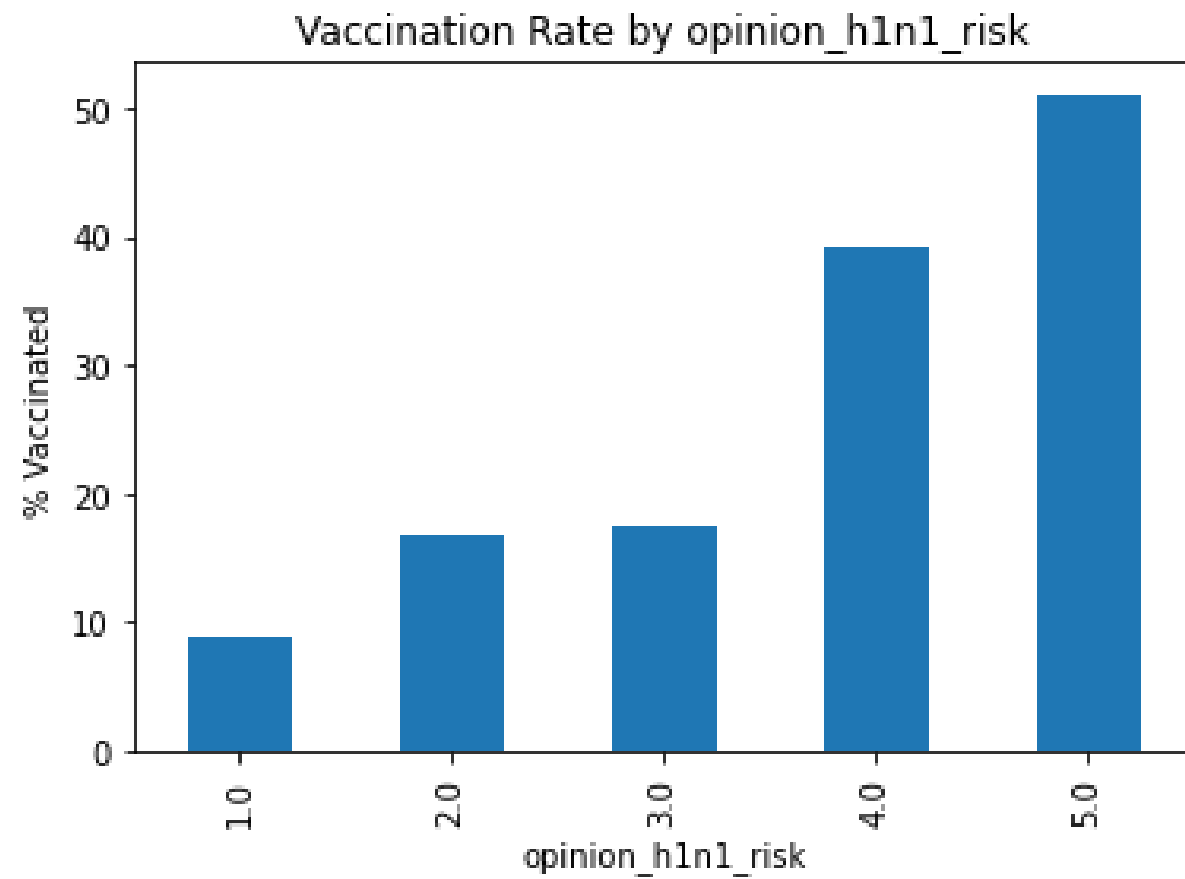
Visualizations

1. Vaccination Rate by Opinion on Vaccine Effectiveness
2. Vaccination Rate by Opinion on H1N1 Risk
3. Vaccination Rate by Doctor's Recommendation
4. Vaccination Rate: Doctor's Rec x Opinion on H1N1 Risk
5. Vaccination Rate: Income x Risk (w/,w/o Rec)

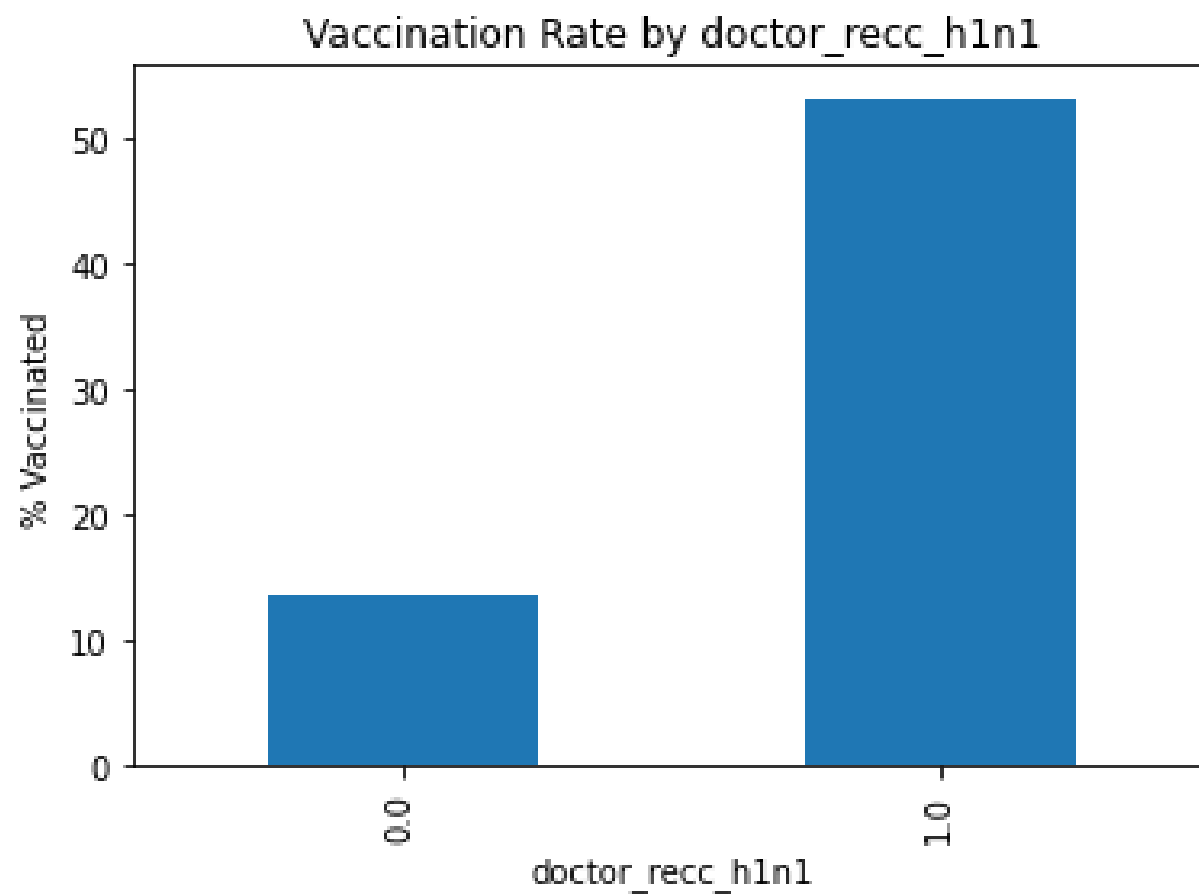
Visual 1



Visual 2

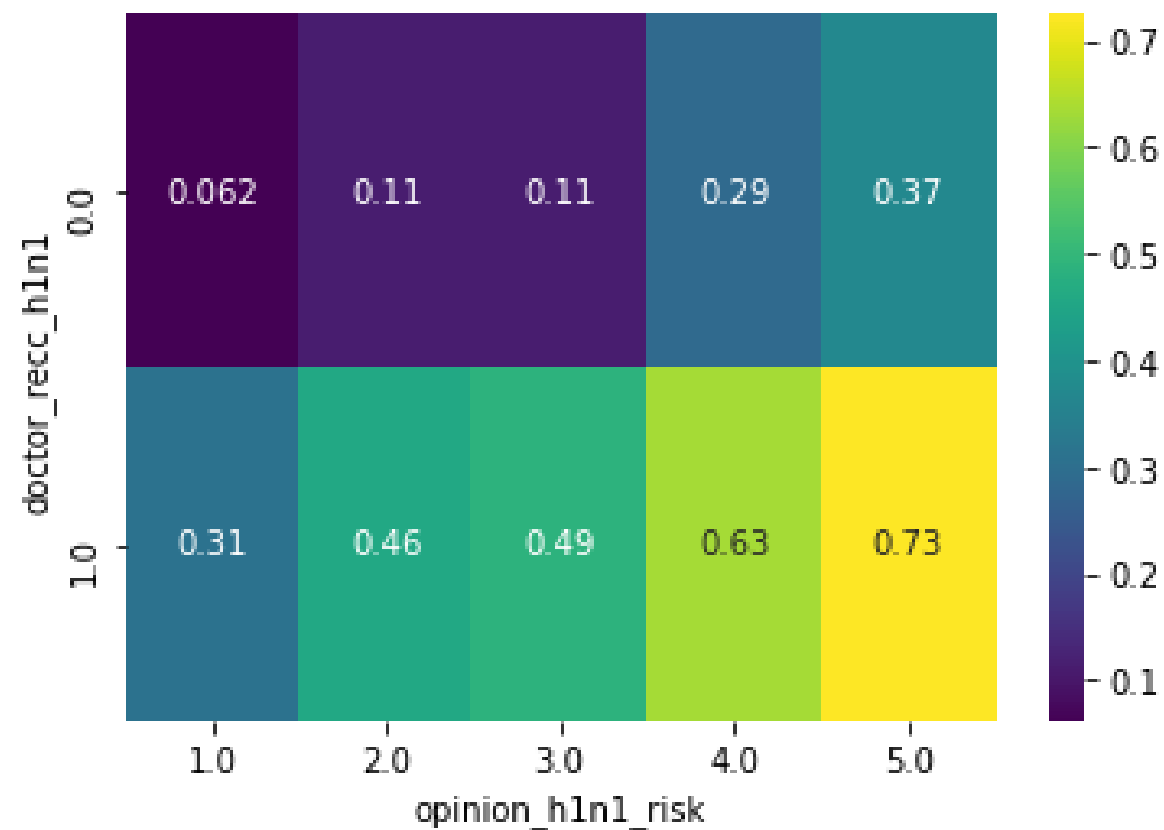


Visual 3

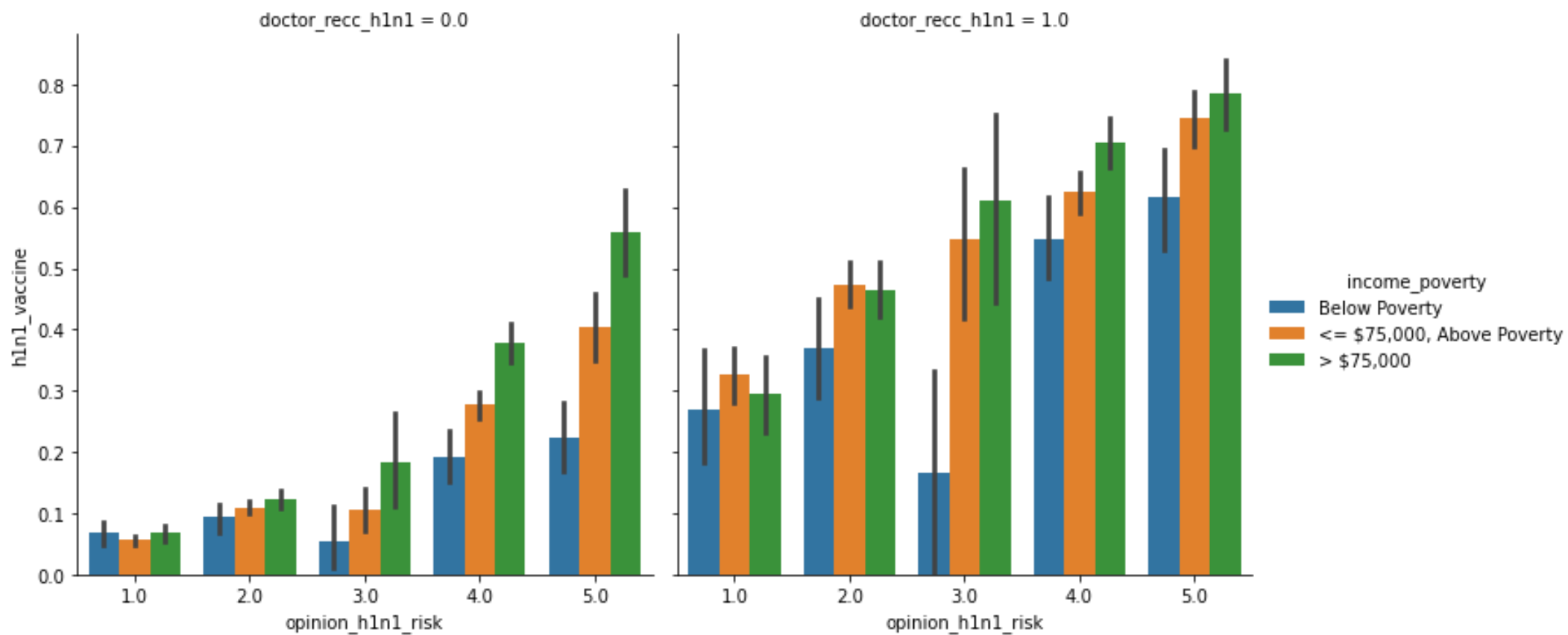


Visual 4

Vaccination Rate: Doctor Recommendation $\times$ Perceived Risk



Visual 5



Findings: The “Doctor Factor”

A doctor’s recommendation is the single strongest predictor of vaccination

Individuals who received a recommendation were significantly more likely to vaccinate, even if they originally thought their personal risk was low

- ▶ Even people who feel at risk don’t always vaccinate — unless advised by a doctor (at **level 5 perceived risk**, *37%* rate *without doctor’s rec* vs. *73% with doctor’s rec*)

Findings: Mindset Matters

- ▶ **Effectiveness:** People who believe the vaccine works are nearly **8x** more likely to get it than skeptics (40% vs 5%)
- ▶ **Risk:** If people don't feel they are in danger of getting sick, they don't get the shot. (rate of 10% at risk level 1 vs. 50% at risk level 5)

Findings: Socio-economic Barriers

- ▶ High-income individuals who feel "at risk" vaccinate at higher rates than low-income individuals who feel the same level of risk (esp. when the latter do not have a doctor's rec)
- ▶ **Black** respondents and the **45–54** age group also showed slightly **lower vaccination likelihoods** in this dataset

Hence, It's not just about "wanting" the vaccine; it's about access and systemic trust.

Other factors

- ▶ Certain employment environment had significant correlation with vaccination status
- ▶ 30% of individuals who reported wearing masks tended to be vaccinated, while 20% of those who didn't were vaccinated
- ▶ 22% of individuals who reported hand-washing tended to be vaccinated, while 14% of those who didn't were vaccinated

The Model

- ▶ Two model types were tested and the winner was Logistic Regression
- ▶ The model achieved an **ROC-AUC of 0.83**
- ▶ In English; on a scale of 0.5 (guessing) to 1.0 (perfect), 0.83 means the model is very strong at distinguishing between those who will and won't get vaccinated

Recommendations

- ▶ Use doctors to promote vaccines
- ▶ Run campaigns specifically targeting *how* the vaccine works to build trust in vaccine effectiveness
- ▶ Outline the likelihood (and severity) of infection during campaigns
- ▶ Target lower-uptake demographic groups (e.g., Black communities and lower-income households) to bridge the trust gap and address systemic barriers

Potential Impact

- ▶ Higher vaccination rates
- ▶ Better targeted campaigns
- ▶ More efficient resource use

Don't be like RFK Jr — Questions?

 **Dr. Mike Varshavski** 
[@RealDoctorMike](#)

An example of why I don't want RFK Jr. in charge of healthcare in the U.S.
[#PolioVaccine](#)



Dr. Mike Varshavski 
[@RealDoctorMike](#)

It took 2 children dying from measles for RFK Jr. to say "the MMR vaccine is the best way to stop the spread of measles."

This didn't need to happen. When he said he convinces parents to not vaccinate their children, that should've told us everything we needed to know. But here we are 😞

Secretary Kennedy   [@SecKennedy](#) · Apr 6

I came to Gaines County, Texas, today to comfort the Hildebrand family after the loss of their 8-year-old daughter Daisy. I got to know the family of 6-year-old Kayley Fehr after she passed away in February. I also developed bonds with and deep affection for other members of

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